

Multivariable Calculus (MT1008)

BSAI, BSDS, BSSE (All Sections)

Course Instructors

Muhammad Ali, Sara Aziz, Sehrish Hassan Shigri

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 70

Total Questions: 4

Date: April 10, 2026

Roll No

Section

Student Signature

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Attempt all the questions

[CLO #2: Analyze and solve advanced mathematical problems by integrating multiple techniques of multivariable calculus, such as optimization, and line/surface integrals.]

Question 1

[7 Marks]

- (a) For a differentiable convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, any point where $\nabla f(x) = 0$ is a global minimizer of f . Is this always true? (1)
- (b) A machine learning system's performance score is determined by tuning two parameters, x and y , according to the function

$$P(x, y) = 4x - x^2 - y^2$$

Due to computational resource limits, these parameters are restricted to the closed domain:

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 5\}.$$

Find the parameter values (x, y) that yield the absolute maximum performance score within D . (6)

[CLO # 3: Evaluate application-oriented problems by applying multivariable calculus techniques such as gradient descent and computation graphs in the relevant physical or computational contexts]

Question 2

[19 Marks]

- (a) Compare forward differentiation and backpropagation in the context of neural network training. Which approach yields better computational efficiency, and why? (2)
- (b) A tiny neural network has one hidden neuron with the following parameters:
Input $x = 2$, weight $w = 1.5$, bias $b = 0.5$, and activation function

$$f(z) = \max(0, z) \quad (\text{ReLU}).$$

The output is:

$$\hat{y} = f(wx + b).$$

Target value is $y = 2$, and the objective is:

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

- (i) Perform the forward pass: compute z , $\hat{y}$, and L . (3)
- (ii) Using backpropagation, compute $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$. (4)
- (iii) Perform one gradient descent update on w using learning rate $\eta = 0.1$. (4)
- (c) Under what conditions on the loss function does gradient descent always converge to a global minimum? (2)
- (d) Discuss the impact of the learning rate on the convergence of a model. What specific problems arise during training if the learning rate is chosen to be excessively large versus excessively small? (2)
- (e) Briefly contrast how Batch Gradient Descent and Stochastic Gradient Descent differ in computing their updates to find a minimum. (2)

[CLO # 3: Evaluate application-oriented problems by applying multivariable calculus techniques such as gradient descent and computation graphs in the relevant physical or computational contexts]

Question 3

[20 Marks]

- (a) Given the gradient at step t : $\nabla f = 5$, velocity $v_t = 2$, learning rate 0.1, and momentum coefficient 0.9.
 - (i) In Nesterov Accelerated Gradient (NAG), the gradient is computed at a look-ahead position rather than the current parameter state. Write the mathematical expression for this position and compute its value for the given parameters. (3)
 - (ii) Compute the new velocity using the Momentum method. (3)
- (b) Explain how the momentum method differs from standard gradient descent, and state one key advantage of NAG over the standard momentum approach. (2)
- (c) Consider two optimization settings:
Case 1: Learning rate= 0.1, momentum coefficient= 0.2
Case 2: Learning rate= 0.5, momentum coefficient= 0.9
Compare the effect of the learning rate and momentum coefficient on the update step, and briefly justify which of the two cases is more likely to cause overshooting or instability. (2)
- (d) A software engineer is optimizing parameters of a system using Newton's method. The objective function is a convex quadratic:

$$L(w) = 2w_1^2 + 0.5w_2^2$$

with initial point $w^{(0)} = (10, 10)$.

- (i) Compute the exact Newton update: (6)

$$w^{(1)} = w^{(0)} - H^{-1} \nabla L(w^{(0)}).$$

- (ii) Notice that the $w^{(1)}$ you calculated is the exact global minimum of the function. Explain why Newton's method converges in a single step for this specific type of objective function. (2)
- (iii) In practical, large-scale optimization systems, standard Newton's method often faces issues with instability or computational infeasibility. Explain the primary reasons why this standard method might fail. (2)

[CLO #2: Analyze and solve advanced mathematical problems by integrating multiple techniques of multivariable calculus, such as optimization, and line/surface integrals.]

Question 4

[24 Marks]

- (a) A cloud system minimizes cost:

$$f(x, y) = x^2 + 4y^2 - 4x + 8y$$

subject to the constraint:

$$x + 2y = 3$$

- (i) Set up the Lagrangian and find the critical points. (8)
 - (ii) Use the bordered Hessian matrix to verify that the critical point corresponds to a local minimum. (4)
- (b) A deep learning system assigns two non-negative weights w_1 and w_2 to control feature activation strength while keeping the total activation energy minimal:

$$\min_{w_1, w_2} w_1^2 + w_2^2$$

subject to:

$$w_1 + w_2 \geq 4, \quad w_1 \geq 0, \quad w_2 \geq 0.$$

- (i) Convert all constraints into standard form $g_i(w) \leq 0$ and construct the Lagrangian using the Lagrange multipliers λ_1 , λ_2 and λ_3 . (2)
- (ii) Write the KKT conditions: (Stationarity, Primal feasibility, Dual feasibility, Complementary slackness). (4)
- (iii) Consider the case where the constraint $w_1 + w_2 \geq 4$ is active ($\lambda_1 > 0$), while the constraints $w_1 \geq 0$ and $w_2 \geq 0$ remain inactive ($\lambda_2 = \lambda_3 = 0$). Under this specific case, solve for the optimal values of w_1 and w_2 . (Note: You do not need to analyze or solve for any other cases.) (5)
- (iv) Explain the role of complementary slackness in KKT conditions. Why does it help in identifying which constraints are active or inactive at the optimum? (1)

Good Luck